Zhisheng Zheng

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Research Profile

I am a Ph.D. student in Computer Science at The University of Texas at Austin, advised by David Harwath. My research focuses on speech and audio generation and understanding. I am particularly interested in controlling how speech is expressed, improving audio and music generation through reinforcement learning, and enabling language models to reason about sound.

Research Interests

Controllable Speech Generation and Translation. Multilingual synthesis, voice cloning, editing, and fine-grained prosodic and duration control (*CtrlSpeech* [3], *VoiceCraft-X* [5], *VoiceStar* [4]); timeline-preserving speech-to-speech translation (*TransDubber* [2]).

Reinforcement Learning for Audio/Music Generation. Reinforcement learning and pairwise reward modeling for generation (*DuetJudge* [1]).

Spatial Audio Understanding and Reasoning. Connecting acoustic perception with large language models for spatial sound understanding (*BAT* [7]).

Self-Supervised Learning. Speech and emotion representations, cost-effective data selection (*emotion2vec* [8], *MT4SSL* [9], and *Unsupervised Active Learning* [10].

Education

The University of Texas at Austin, Austin, USA — Aug 2024 – Present
Ph.D. Student, Computer Science
Advisor: Prof. David Harwath

Shanghai Jiao Tong University, Shanghai, China — Sep 2020 – Jun 2024
B.E. in Information Engineering; Zhiyuan Honors Program of Engineering
Advisor: Prof. Xie Chen

Research Experience

NVIDIA — Feb 2026 – Dec 2026, Santa Clara, USA
Applied Deep Learning PhD Research Intern

Amazon — May 2025 – Feb 2026, Seattle, USA
Applied Scientist Intern

Microsoft Research Asia — Apr 2024 – Aug 2024, Beijing, China
Research Intern (Supervisor: Lei He)

The University of Texas at Austin, SALT Lab — Jul 2023 – Jan 2024, Austin, USA
Visiting Scholar (Supervisors: David Harwath, Eunsol Choi)

Shanghai Jiao Tong University, X-Lance Lab — Dec 2021 – Jun 2024, Shanghai, China
Research Intern (Supervisor: Xie Chen)

Preprints

[1] Reinforcement Learning for Music Generation with Pairwise Reward Modeling.
Zhisheng Zheng, Zhifeng Kong, Sang-gil Lee, Jaehyeon Kim, Arushi Goel, David Harwath, Mohammad Shoeybi, Bryan Catanzaro, Wei Ping
Submitted to ICLR 2027.

[2] Learning Timeline-Preserving Speech-to-Speech Translation.
Zhisheng Zheng, David Harwath

Selected Publications

[3] CtrlSpeech: Coarse-to-Fine Control for Expressive Speech Synthesis.
Zhisheng Zheng, Xiaohang Sun, Zhu Liu, Caren Chen, Rohith Kumar, Manoj Aggarwal, Gerard Medioni, David Harwath
Interspeech, 2026 (***Oral***).

[4] VoiceStar: Robust Zero-Shot Autoregressive TTS with Duration Control and Extrapolation.
Puyuan Peng, **Zhisheng Zheng**, Shang-Wen Li, Abdelrahman Mohamed, David Harwath

Findings of ACL, 2026.

[5] VoiceCraft-X: Unifying Multilingual, Voice-Cloning Speech Synthesis and Speech Editing.
Zhisheng Zheng, Puyuan Peng, Anuj Diwan, Cong Phuoc Huynh, Xiaohang Sun, Zhu Liu, Vimal Bhat, David Harwath
EMNLP, 2025.

[6] Scaling Rich Style-Prompted Text-to-Speech Datasets.
Anuj Diwan, **Zhisheng Zheng**, David Harwath, Eunsol Choi
EMNLP, 2025.

[7] BAT: Learning to Reason about Spatial Sounds with Large Language Models.
Zhisheng Zheng, Puyuan Peng, Ziyang Ma, Xie Chen, Eunsol Choi, David Harwath
ICML, 2024.

[8] Emotion2vec: Self-Supervised Pre-Training for Speech Emotion Representation.
Ziyang Ma, **Zhisheng Zheng**, Jiaxin Ye, Jinchao Li, Zhifu Gao, Shiliang Zhang, Xie Chen
Findings of ACL, 2024.

[9] MT4SSL: Boosting Self-Supervised Speech Representation Learning by Integrating Multiple Targets.
Ziyang Ma, **Zhisheng Zheng**, Changli Tang, Yujin Wang, Xie Chen
Interspeech, 2023 (***Best Student Paper Shortlist***).

[10] Unsupervised Active Learning: Optimizing Labeling Cost-Effectiveness for Automatic Speech Recognition.
Zhisheng Zheng, Ziyang Ma, Yu Wang, Xie Chen
Interspeech, 2023.

Selected Awards & Honors

Shanghai Outstanding Graduates 2024

SenseTime Scholarship for Undergraduate AI Researchers
30 winners nationwide each year, SenseTime 2023

Rongchang Science and Technology Innovation Scholarship (<0.1%)
Shanghai Rongchang Public Welfare Foundation 2023

Best Student Paper Shortlist, Interspeech 2023

Tencent Scholarship (Top 2%)
Tencent Technology (Shenzhen) Co., Ltd. 2021

Zhiyuan College Honors Scholarship (Top 10%), Zhiyuan College 2020–2024

Academic Service

Conference Reviewer / Program Committee:
ICLR 2026–2027, ICML 2025–2026, ACL Rolling Review 2024–2026, AAAI 2026